RAJALAKSHMI ENGINEERING COLLEGE

Vijay Karthick S 251001301-2025-IT

V2

Started on	Wednesday, 7 October 2026, 9:07 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:08 AM
Time taken	32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    long long diff = (*(long long*)a - *(long long*)b);
    return (diff > 0) - (diff < 0);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    long long *arr = (long long*)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        if (scanf("%lld", &arr[i]) != 1) {
            free(arr);
            return 0;
        }
    }

    qsort(arr, n, sizeof(long long), compare);

    long long max_sum = 0;
    for (int i = 0; i < n; i++) {
        max_sum += arr[i] * i;
    }

    printf("%lld\n", max_sum);

    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course